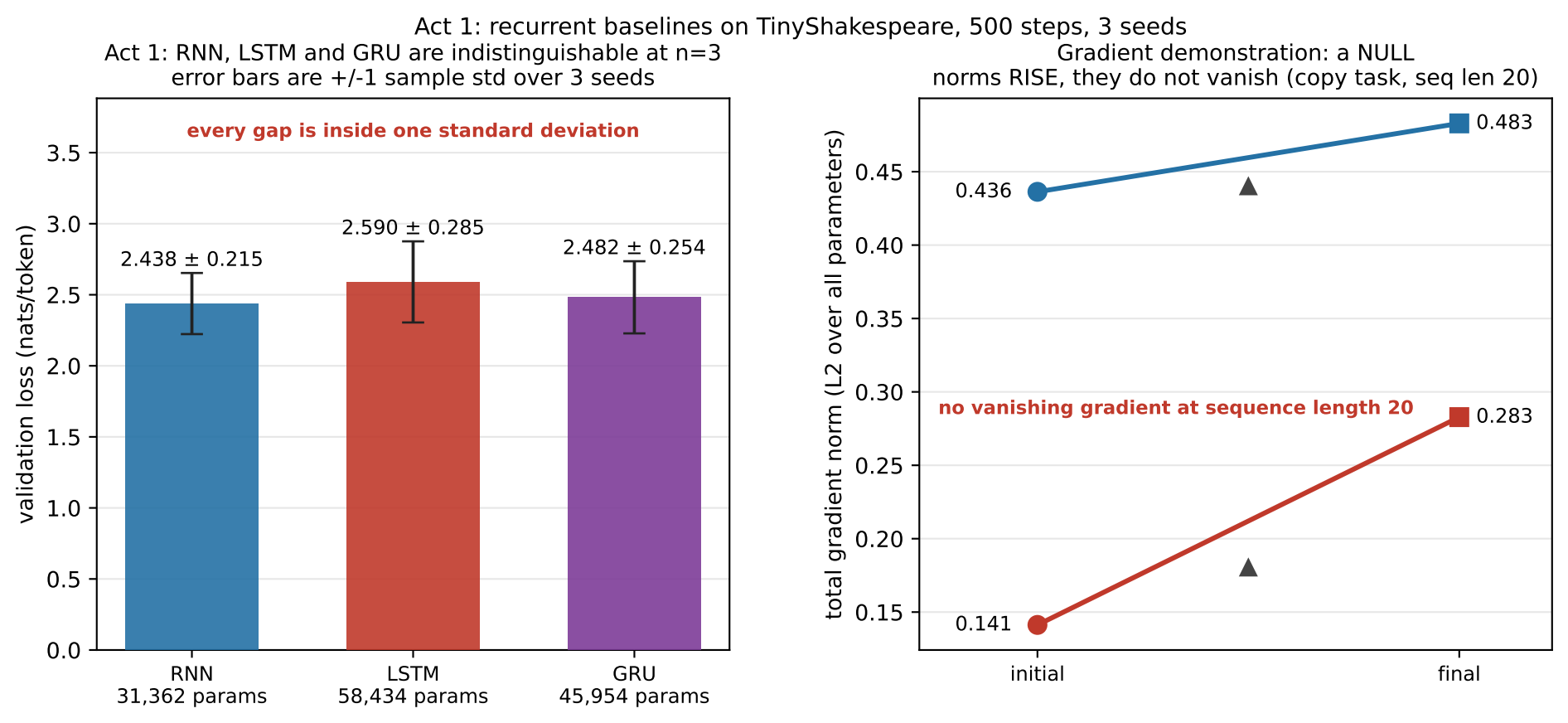
Act 1: recurrent baselines on TinyShakespeare, 500 steps, 3 seeds
Act 1: RNN, LSTM and GRU are indistinguishable at n=3
error bars are +/-1 sample std over 3 seeds
every gap is inside one standard deviation
validation loss (nats/token)
2.438 ± 0.215
2.590 ± 0.285
2.482 ± 0.254
RNN
31,362 params
LSTM
58,434 params
GRU
45,954 params
Gradient demonstration: a NULL
norms RISE, they do not vanish (copy task, seq len 20)
total gradient norm (L2 over all parameters)
0.436
0.483
no vanishing gradient at sequence length 20
0.141
0.283
initial
final